Lecture 17

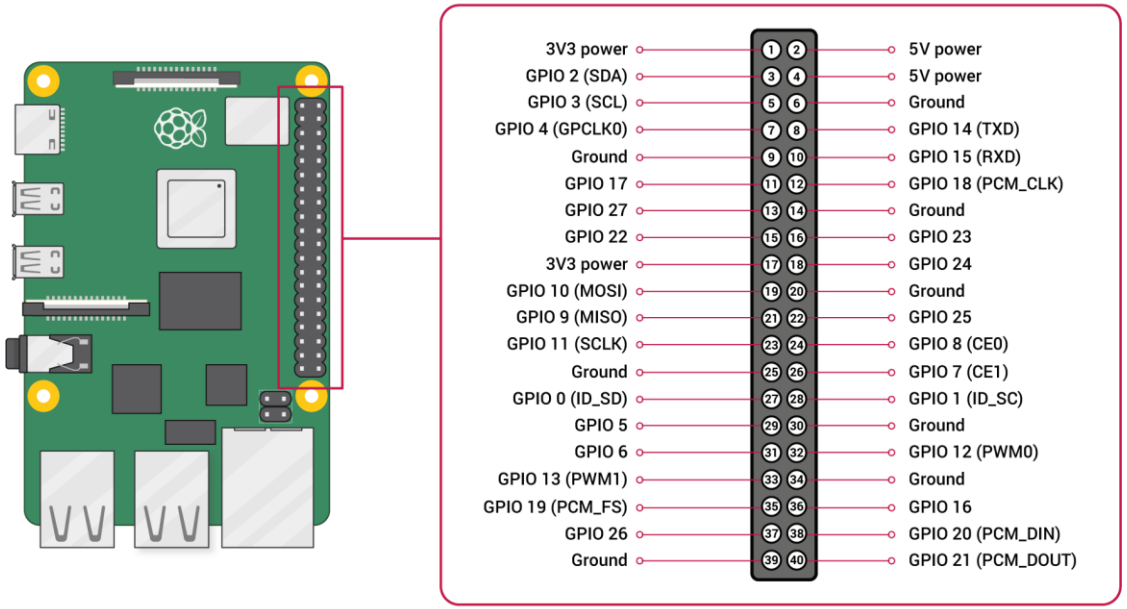
Lab 3: Introduction and Examples

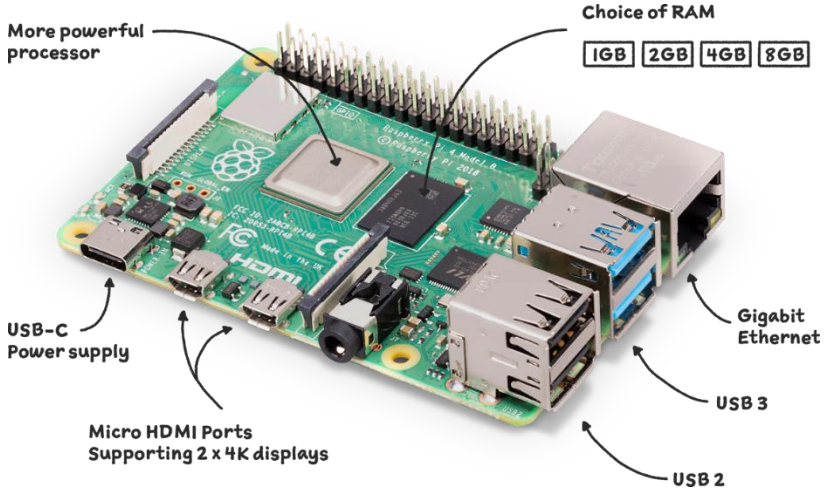
Prof. Deming Chen

Slides initially made by previous TA Junhao Pan

Track 1: IoT System

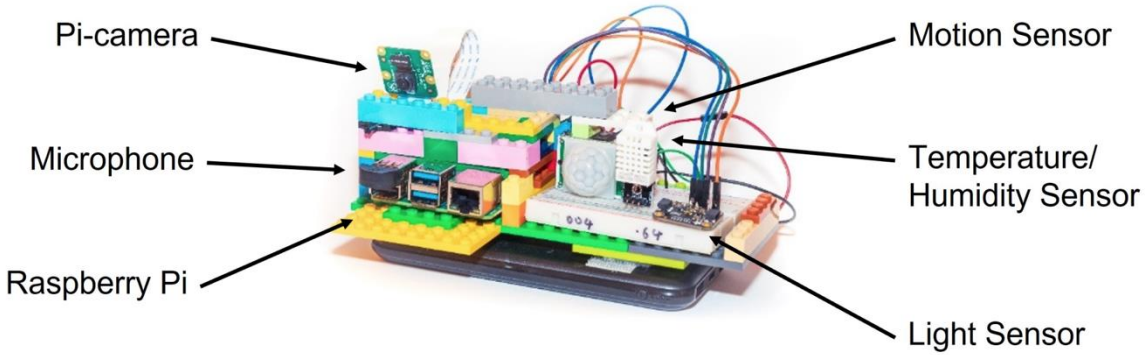
- Based on Raspberry Pi + Coral accelerator
- Extra hardware is allowed
  - USB Peripherals (microphone, audio adaptor, gamepad, etc.)
  - GPIO interface (assorted sensors, LED modules, actuator/buttons, servo motors, etc.)
  - cameras (PiCamera, infrared/depth cameras, etc.)
  - Portable battery charger





Track 1: Example 1—Smart Home system

- Prototype device: Raspberry Pi with assorted sensors
  - Breadboard, Lego, soldering, etc.
  - Coral TPU
- Edge nodes deployed in household environments
  - Collect temperature & humidity, light, motion, sound, and RGB videos
  - Include data for multiple users with interactions

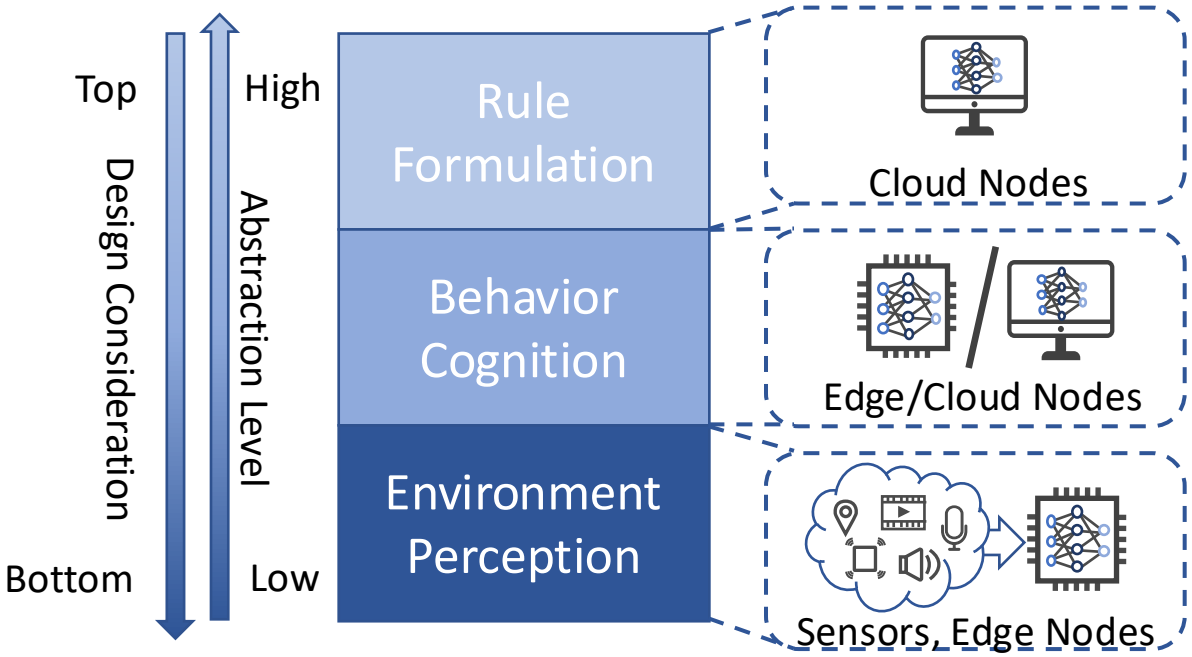


	Temperature	Humidity	Lux	Infrared	Visible	Full Spectrum	Motion
2022.07.22 17:33:40	27.1	51.9	85.54944	1245	81593641	81594886	0
2022.07.22 17:33:41	27.1	51.9	85.22304	1245	81593639	81594884	1
2022.07.22 17:33:42	27.1	52	84.55066	1253	82117928	82119181	1
2022.07.22 17:33:43	27.1	52	83.20589	1269	83166506	83167775	0
…	…	…	…	…	…	…	…

Recorded Event:
{'User': 101, 102, 111, 112, 'Activity': Eat}

Junhao Pan, Zehua Yuan, Xiaofan Zhang, and Deming Chen, "YouHome System and Dataset: Making Your Home Know You Better", *Proceedings of IEEE International Symposium on Smart Electronic Systems*, December 2022.
Zehua Yuan, Junhao Pan, Xiaofan Zhang, and Deming Chen, "HomeSGN: A Smarter Home with Novel Rule Mining Enabled by a Scorer-Generator GAN", Proceedings of IEEE/ACM Asia and South Pacific Design Automation Conference, January 2024.

Track 1: Example 1—Smart Home system



- Environment perception
  - Edge nodes embedded in the targeted environment, equipped with multiple sensors and lightweight computing units.
- Behavioral cognition
  - Neural networks or other ML algorithms that can identify users' identities and classify their activities after adequate training to understand the physical world.
- Rule formation
  - A layer that produces the final outcome by formulating rules from the observed patterns of the interactions of the residents with their home

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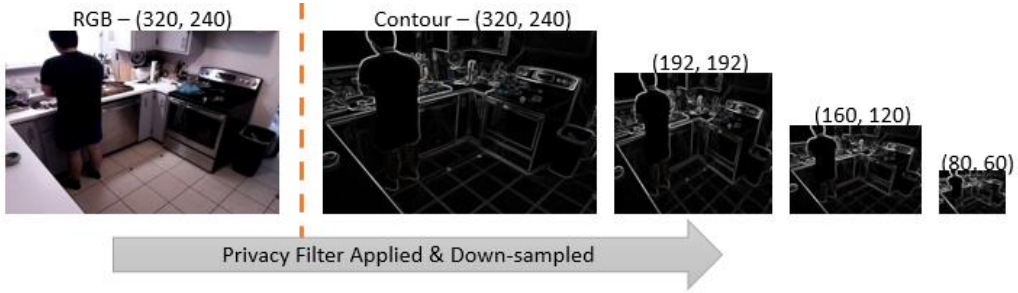
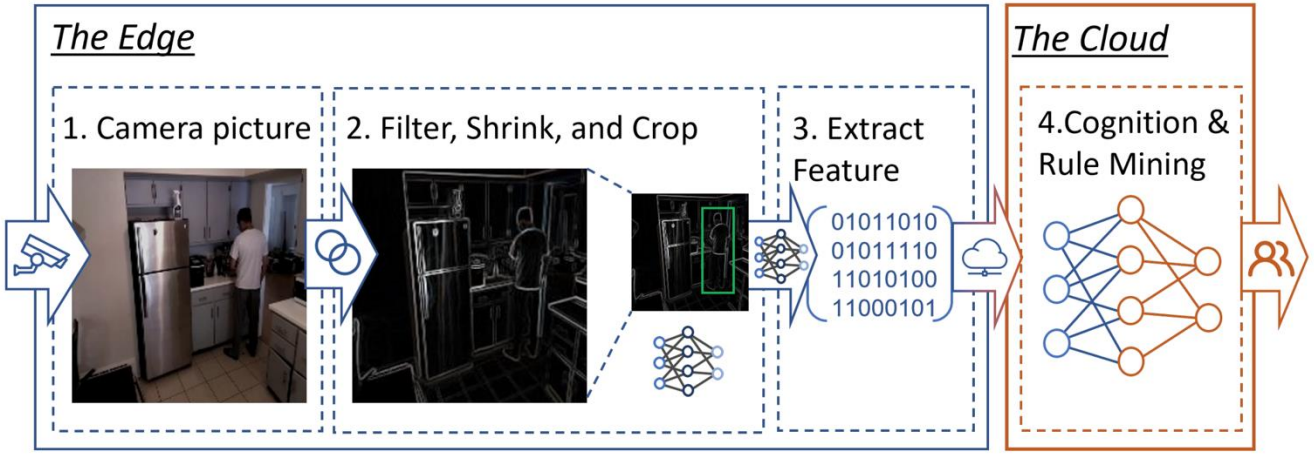
Track 1: Example 1—Smart Home system

- System privacy:
  - Distributed Model Deployment (DMD): separate NN models between the cloud and the edge
  - Privacy Filter and Low-Resolution Images: Sobel filters and Gaussian filters
- Activity cognition:
  - User Identification: Human Re-Identification network models
  - Event Recognition: Classification models
  - Realtime ML inferencing on Coral TPU
- Rule mining:
  - Discover frequency patterns
  - Formulate and propose beneficial rules





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Track 1: Example 1—Smart Home system

- Development and Deployment
  - Development: Most of the coding, training, integration, etc.
  - Deployment: Consider real-life use cases, practicality, maintainability, robustness, etc.

- Development on PC/Server
  - Data collection, processing pipeline
  - Neural networks training—**PyTorch, TensorFlow**
  - Flow verification

- Deployment on Edge platform:
  - PiCamera, other sensor drivers
  - TF-lite—Coral TPU—Achieve real-time inference speed

- YouHome Dataset and Tutorial
  - GitHub repo: [GitHub - UIUC-ChenLab/YouHome-Dataset: This is the repository of YouHome Activities of Daily Living (ADL) Dataset]
  - Great opportunity to learn PyTorch

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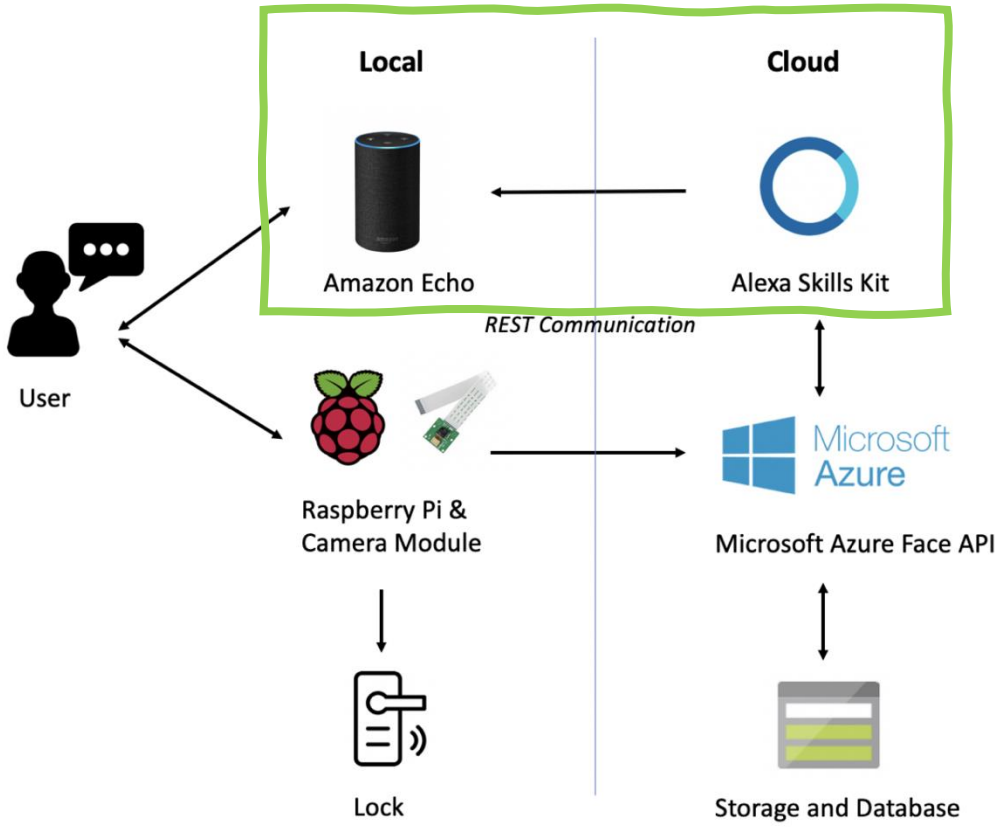
Track 1: Example 2—Smart Lock with facial recognition

- Interact with Alexa: Intent, Utterance, and Slots
  - Define ***intent***:
    - *PlanTrip*
  - Give example ***utterance:***
    - *I am going on a trip on Friday …*
    - *I want to visit Seattle …*
    - *I will drive to Chicago to have dinner …*
  - Identify **slots**
    - *I am going on a trip on Friday …*
    - *I want to visit Seattle …*
    - *I will drive to Chicago to have dinner …*

- Perform an Alexa **skill**
  - Alexa tries to understand your **utterance** and translate it to an **intent**
  - Each **skill** corresponds to an action



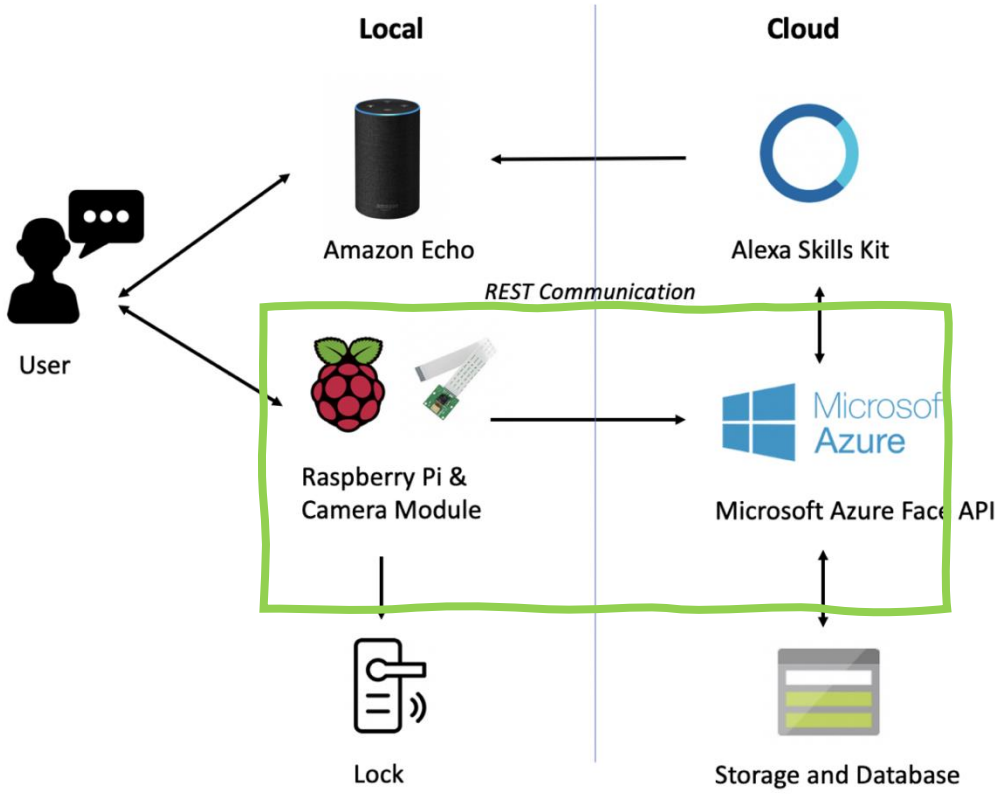
[Create Intents, Utterances, and Slots | Alexa Skills Kit (amazon.com)]

Track 1: Example 2—Smart Lock with facial recognition

- (Always) Listen to Alexa
  - Local script (on RPi) to monitor if a skill is invoked

- Edge actions:
  - Take a picture with PiCamera
  - Upload the picture to Azure
  - Get recognition results from Azure
  - Send a signal to lock if unlock

- Cloud actions:
  - Analyze the uploaded picture
  - Identify whether the face is registered
  - Return results to RPi



Raspberry Pi Example Projects

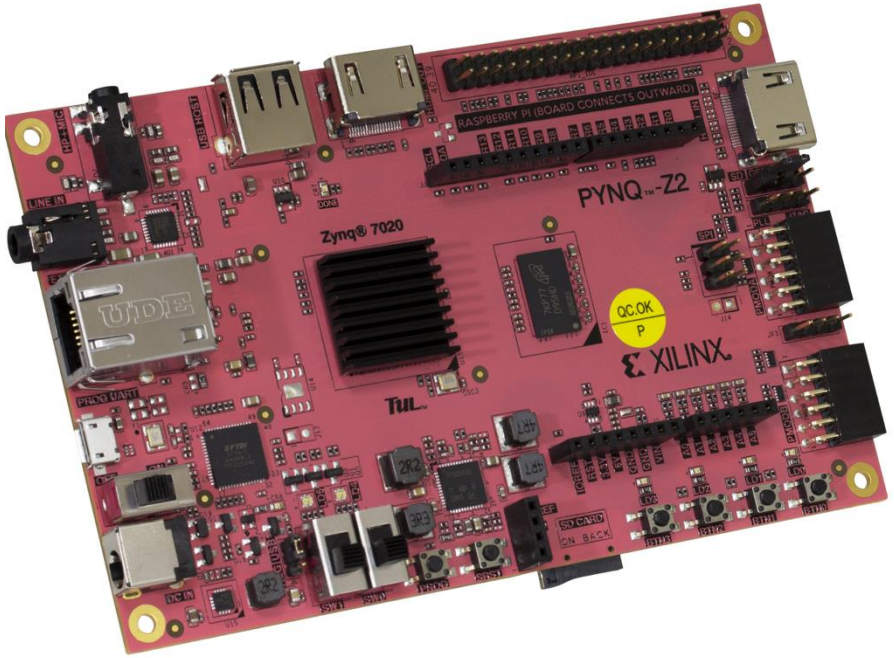
https://www.youtube.com/watch?v=7litEjoVu_s

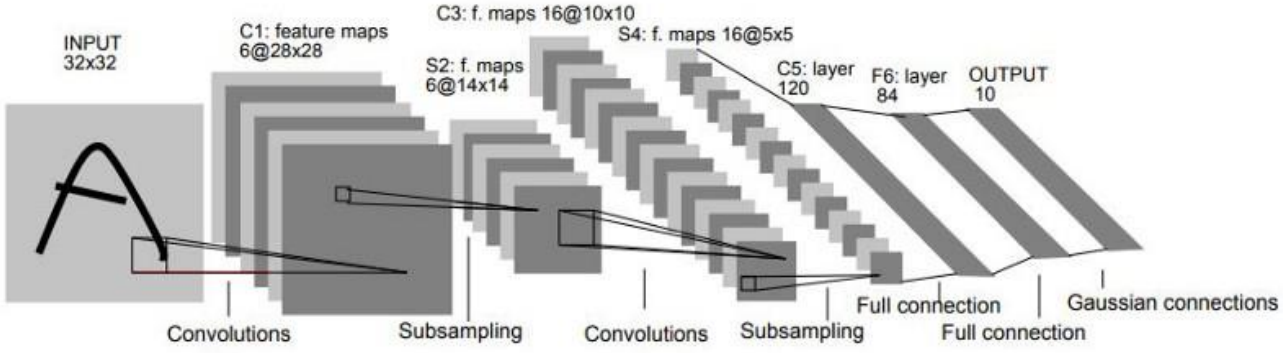
Track 2: DNN Accelerator and DNN Acceleration Library

- Based on DNN accelerators, computing platforms, or ML algorithms
- Hardware approach:
  - Explore new hardware platforms
  - Design and deploy novel accelerators
- Software approach:
  - Design and optimize novel machine learning algorithms
- HW-SW Co-design is even better!

Track 2: Example 1—Accelerate DNN on FPGA

- LeNet-5: a simple and straightforward CNN to classify handwritten characters (MNIST)
- Architecture:
  - Convolution, kernel=5*5, no padding
  - Average pooling, pool=2*2, stride=2
  - Convolution, kernel=5*5, no padding
  - Average pooling, pool=2*2, stride=2
  - Fully connected layer, 400*120
  - Fully connected layer, 120*84
  - Fully connected layer, 84*10
- Easy to deploy on embedded FPGA





Track 2: Example 1—Accelerate DNN on FPGA

```
// Convolution Layer 3
void convolution3(float input[6][14][14], float weights[16][6][5][5], float bias[16],
float output[16][10][10])
{
    for(int co = 0; co < 16; co++)
        for(int h = 0; h < 10; h++)
            for(int w = 0; w < 10; w++)
            {
                float sum = 0;
                for(int i = h, m = 0; i < (h+5); i++, m++)
                {
                    for(int j = w, n = 0; j < (w+5); j++, n++)
                        for (int ci = 0; ci < 6; ci++)
                            sum += weights[co][ci][m][n] * input[ci][i][j];
                }
                output[co][h][w] = sum + bias[co];
            }
}
```

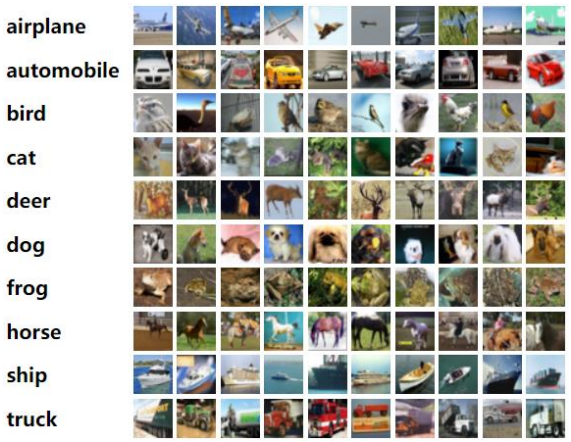
- Potential Approaches:
  - Aggressive quantization—retraining
  - Explore parallelism—pipelining, partitioning, dataflow, etc.
  - More complex network architecture
- Do mini-lab to learn VivadoHLS

Track 2: Example 2—Accelerate DNN on GPU

- Google Colab provides a free platform for running Python on Jupyter notebooks
  - Colab provides access to free GPUs
  - To access GPU in Colab
    - Runtime -> Change runtime type -> Hardware accelerator -> Choose GPU
    - Edit -> Notebook Settings -> Hardware accelerator -> Choose GPU

- Using GPU in Colab can significantly speed up your computations

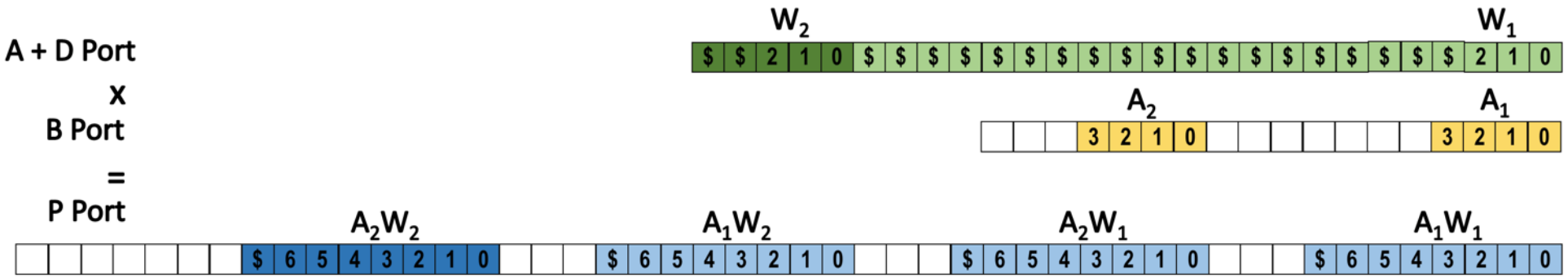
- Example (GPU mini-lab): Training neural network on Cifar-10
  - Tensorflow ML framework
  - Google Colab notebooks
  - Free GPU
  - Cifar-10 Dataset: a collection of 60,000 32x32 color images in 10 classes, with 6,000 images per class
  - At least 85% accuracy (current SOTA: 99.5%)



Track 2: Example 3—Extend existing computing library

- HiKonv: maximizes the convolution throughput of a processing unit
  - Benefits low-bitwidth quantized data
  - Appliable to arbitrary bitwidth multiplier
  - Features bit-wise parallel computation

- Intuition: INT4 on Xilinx DSP48E2

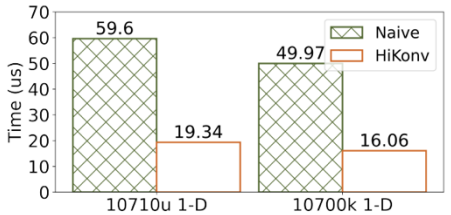


Xinheng Liu, Yao Chen, Prakhar Ganesh, Junhao Pan, Jinjun Xiong, and Deming Chen, "HiKonv: High Throughput Quantized Convolution with Novel Bit-wise Management and Computation", *Proceedings of IEEE/ACM Asia and South Pacific Design Automation Conference*, January 2022.

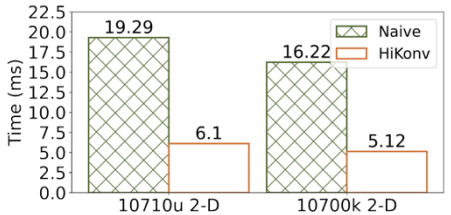
Track 2: Example 3—Extend existing computing library

- Advantages of HiKonv:
  - Highest possible throughput for convolution on any multiplier
  - Reduced latency, almost reaching theoretical speedup

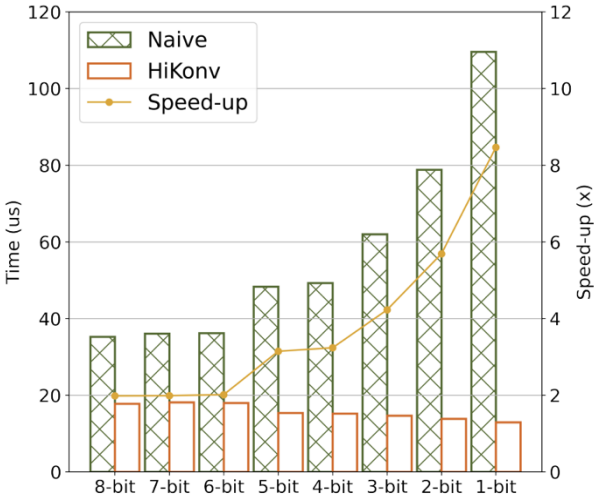
- Possible approach:
  - Evaluate and explore possible improvements on ARM (with ARM compute library)
  - Try to apply the similar concept to matrix multiplication or other operations



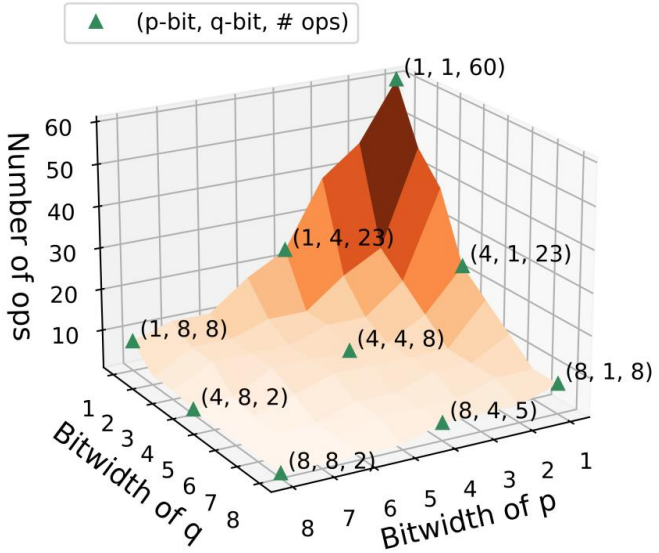
(a) 1-D Convolution

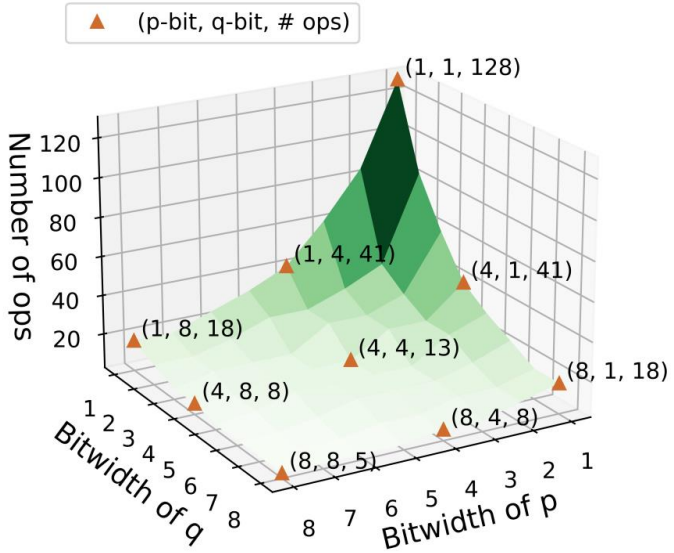


(b) 2-D Convolution



(c) Speedup for different bitwidths.





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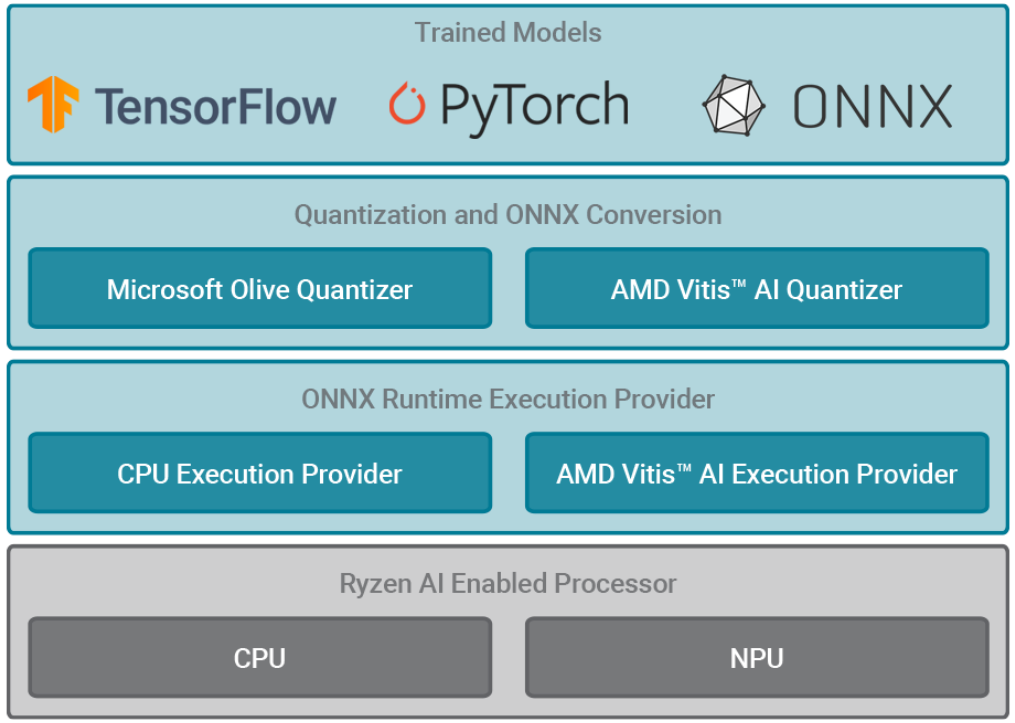
Track 2: Example 4—Explore AI Capability on AIE

- AMD Ryzen AI: embedded Neural Processing Unit (NPU) for **AI inference** on Ryzen PCs
  - Fast and power-efficient
- Supports PyTorch and TensorFlow through [Ryzen AI Software]
  - Train: PyTorch, TF, ONNX models on server/cloud
  - Quantize: quantize to INT8, save to ONNX model
  - Deploy: manages and runs the model on AI hardware
- Pretrained models are available in a model zoo: quick and easy development
- Supported CPUs:
  - 7940HS, 7840HS, 7840U, 7640HS, 7640U
  - 8945HS, 8845HS, 8840HS, 8840U, 8645HS, 8640HS, 8640U
  - HX 375, HX 370, …



Proposal

- Due 3/29 Sunday Midnight

- 1 to 2 pages to include 4 paragraphs with illustrations
  1. Which track? IoT or Computing?
  2. Overall description of your system/algorithm/methodology. State your novelty. What is different?
  3. Key features and difficulty level (please justify)?
  4. What do you expect in your midpoint and final results?

- Submit by Sunday (3/29), discuss with your TA next Monday
  - Monday attendance is required

Midpoint, demos, and in-class presentations

- 4/20—midpoint demo, 5/4—final demo
- Midpoint demo is similar to the final demo:
  - Completeness, Quality, Difficulty, Novelty
  - TA checks your progress and preliminary results if any
  - Adjust goals only if necessary
- Final demo is the grand finale of your lab
  - Make sure everything works!
  - Start your report early
- In-class presentations: 4/23 to 4/30
  - Presentation schedule to be released soon
  - Show off to the entire class: learn from as well as educate your fellow students
  - Receive final feedback from TAs and Professor
  - Each team has 10-min to present, followed by 5-min Q&A. More details next.

Project Presentation—Title: 1 slide

- A cool name for your project
- Names of you and your partner
- Work partition: who is the quarterback, who is the waterboy
- Name of your TA

Project Presentation—Motivation/Problem

- Why do you think your work is important?
- What are the existing solutions?
- What are the challenges?
- What do you do to improve? (on a high level)

Project Presentation—Design and Methodology

- What does your design look like?
  - Explain with graphs/pictures
- What does your design do?
  - Explain each part with some technical details
- What benefit does your design make?
  - Any specific reason you choose to design it this way?

Project Presentation—Preliminary Results

- Any results you wish to present.
- Any short demo?
- Does it show your design is good? Why? How do you improve?
- Does it show your design is bad? Why? How do you adjust?

Lab 3 Conclusion

- Very open-ended project
  - Be creative.
  - Be reasonable.
  - Make yourself proud.
  - Make 479 worth taking.
- TAs are here to help
  - Come to labs and office hours
  - Don’t wait until the last minute
- More details on the document/website/Campuswire